Kelvin's charged-ring molecule

Induction and Kelvin's own Zeeman test of his 1899 ring, with every charge under the Lorentz force. CLAUDE.md §6 (Kelvin); the electron test bench for H13, H21 and H23

kronspice simulation report · 2026-11-07 · 24 charges per half, RK4

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Summary

Lord Kelvin's "Magnetism and Molecular Rotation" (1899) pictures a molecule as a rigid charged ring filled with an oppositely charged fluid that can slide round it. This example builds that ring with standard electrodynamics only, every charge under the Lorentz force, and runs the two things Kelvin did with it: induction when a field is switched on, and his own Zeeman test, which made him give the model up. It is the electron test bench on which *space_vortex* (H13), *kron_electron* (H21) and *photon_electron* (H23) are measured.

- **Induction matches Kelvin and Larmor.** Tube and fluid gain moments of momentum $-/+1.11207\times10^{-10}$ kg·m²/s = $-/+e\Phi/2\pi$, and the fluid turns at Larmor's rate $eB/2w = 4.44829\times10^{-6}$ rad/s.
- **The flux is checked two ways:** 6.98736×10^{-4} Wb by Biot–Savart against 6.98732×10^{-4} Wb from Maxwell's elliptic-integral mutual inductance.
- **The ledger closes:** source work -4.94683×10^{-16} J = kinetic 4.94683×10^{-16} J + mutual-field -9.89365×10^{-16} J, residual -8×10^{-16}.
- **Kelvin's gyrostat as written broadens the Zeeman line** (splitting $(\omega_L/2)\cdot\cos\theta$; ensemble FWHM 0.0188), which is his stated reason for discarding the model.
- **The full ring gives sharp lines** (splitting 0.0243 … 0.0260 at every orientation, FWHM 0.0037), removing that objection, **but still fails Zeeman's test:** its light along B is nearly unpolarised (−0.02 / −0.03 against the measured ±1), and the splitting is set by the total mass w + w′, not by e/m of the electron.

Background

The ring is a non-conducting tube of radius a carrying +e (mass w′), filled with an incompressible fluid carrying −e (mass w) that slides without friction. Kelvin used it in §§1–3 of his paper to describe induction, and in §§5–7 to ask whether such molecules give the Zeeman effect. He discarded it after Zeeman's splitting (p. 357). The standard classical model beside it is Lorentz's bound charge, which gives the normal Zeeman effect because all of its charge has one charge-to-mass ratio (the condition of Larmor's theorem).

The model

- Twenty-four charges per half (tube and fluid), integrated with fourth-order Runge–Kutta (RK4).
- **Induction (Kelvin §§1–3).** A coaxial coil's current is ramped on, held and ramped off. Its field comes from Biot–Savart (*em.Loop*), the induced E from Faraday's law, and the force on each charge from *em.LorentzVector*. The flux is checked against Maxwell's elliptic-integral mutual inductance (*em.MutualCircles*).
- **Zeeman test (Kelvin §§5–7).** The ring is held by isotropic angular springs ($\omega_0 = 1$) in a steady field, kicked, and the spectrum of its motion and of its radiating magnetic moment is taken over twelve orientations. Three models are compared: Lorentz's bound charge, Kelvin's gyrostat as he wrote it (fixed internal spin along the axis), and the full ring (fluid free to slip, rigid-body rotation with *hyper/quat*, nothing linearised).

Results

Part 1: induction

Ring a = 5 cm, ±1 μC, tube and fluid 10 g each; coil radius 10 cm, 5 cm below, 20 kA·turns, 1 s ramps.

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quantity	simulated	expected
flux through the ring	6.98736×10^{-4} Wb (Biot–Savart)	6.98732×10^{-4} Wb (Maxwell, elliptic)
moment of momentum, tube / fluid	-1.11207×10^{-10} / $+1.11207\times10^{-10}$ kg·m²/s	$-/+e\Phi/2\pi = 1.11207\times10^{-10}$ (Kelvin's corrected eAM/2π)
net angular impulse on the ring	-5.5×10^{-30}	0: the halves cancel; the coil gets none back
fluid angular velocity	4.44829×10^{-6} rad/s	Larmor's $eB/2w = 4.44829\times10^{-6}$ rad/s
ring moment	-1.11×10^{-14} A·m² (diamagnetic), 1.44×10^{-14} N repulsion	Langevin diamagnetism
ledger	source work -4.94683×10^{-16} J = kinetic 4.94683×10^{-16} J + mutual-field -9.89365×10^{-16} J	residual -8×10^{-16}
after switch-off	tube 1.5×10^{-26}, fluid -1.5×10^{-26} kg·m²/s; net source work -5×10^{-32} J	everything at rest

Kelvin's ring is a mechanical picture of Larmor's theorem. At atomic scale (electron charge and mass, 1 T) the same fluid turns at 8.794×10^{10} rad/s.

Part 2: Kelvin's Zeeman test

$\omega_0 = 1$, $\omega_L = eB/2w = 0.05$, 12 orientations, spectral resolution $2\pi/T = 0.0026$. θ is the angle between the ring's axis and B.

model	splitting vs cos θ	ensemble upper line (FWHM)	circular polarisation along B, radiated (upper / lower)	across B: centre parallel, sides perpendicular to B
Lorentz bound charge	0.0513 at every θ	sharp, 0.0038	−1.00 / +1.00	100 % / 100 %
Kelvin gyrostat as written	$(\omega_L/2)\cdot\cos\theta$: 0 … 0.0243	broad, 0.0188	+0.72 / −0.71	67 % / 59 %
Kelvin ring, full Lorentz force	0.0243 … 0.0260 at every θ	sharp, 0.0037	−0.02 / −0.03 (motion −0.47 / +0.46; analytic 5/11)	64 % / 71 %

In the last column, "centre parallel, sides perpendicular" is the share of the centre line polarised parallel to B and of the side lines polarised perpendicular to it, viewed across the field.

- **The gyrostat** reproduces Kelvin's reason for discarding the model (p. 357): with a fixed spin the splitting goes as cos θ, so randomly oriented molecules broaden the line instead of splitting it.
- **The full ring** is new. When the ring tilts, the flux through it changes and the fluid's spin follows, since its canonical angular momentum $w\cdot a^2\cdot u - e\Phi/2\pi$ is conserved (u is the fluid's angular velocity). The gyroscopic vector becomes $G = (ea^2/2)(B - 2(n\cdot B)n)$, the reflection of B in the ring's plane (n is the ring axis), whose length does not depend on orientation (*GyroVector*). Every ring splits by $eB/2(w + w')$ and the lines are sharp, which removes Kelvin's and Larmor's objection.
- **It still fails Zeeman's test.** The modes circle G, not B, so the light from the ring's own moment is nearly unpolarised along the field where Zeeman measured ±1, and the splitting is set by the total mass $w + w'$, not the electron's e/m. A ring of two opposite charges with different charge-to-mass ratios cannot give the normal Zeeman effect; Lorentz's single charge can.

Consequence for the electron test bench

Any toroidal model of the electron must give a splitting set by e/m_e and the observed circular polarisations before the anomalous Zeeman effect, g and spin ½ are even asked for. That is the bar applied to the ring models of H13, H21 and H23.

Validation

- *main_test.go*: induction identities and ledgers; the Zeeman test for all three models; |G| constant.
- Twin check: flux by Biot–Savart against Maxwell's elliptic mutual inductance (6.98736 against 6.98732 $\times10^{-4}$ Wb).
- Energy ledger residual -8×10^{-16} J on 4.94683×10^{-16} J of kinetic energy.

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Limits of the model

- Each half is represented by 24 point charges; the fluid is incompressible and frictionless.
- The Zeeman test uses isotropic angular springs ($\omega_0 = 1$) to hold the ring, as a stand-in for whatever binds a molecule, and dimensionless units.
- The example has no run_output.txt or figures; the numbers above are those recorded in its README.

Reproducing the results

```
go run ./examples/kelvin_ring
go test ./examples/kelvin_ring          # -short skips the Zeeman ensemble
```

Sources: W. Thomson (Lord Kelvin), "Magnetism and Molecular Rotation," Proc. R. Soc. Edinb. 22, 631–635 and Phil. Mag. 48, 236–239 (1899); Mathematical and Physical Papers Vol. V, Art. 175, pp. 354–358; notes in doc/kelvin-references.md. H. A. Lorentz, bound-charge model of the normal Zeeman effect; P. Zeeman, Phil. Mag. 43, 226 (1897); J. Larmor, Larmor's theorem (1897); J. C. Maxwell, Treatise Vol. II, mutual inductance of coaxial circles.

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